

Qn	Suggested Answer
2	
(a)(i)	Acceleration is the <u>rate of change of velocity</u> .
(a)(ii)	The <u>gradient</u> of the tangent to the v - t graph gives the acceleration. Acceleration <u>decreases continuously to zero (or near zero) at 16.0 s</u> .
(a)(iii)	Finding gradient of tangent drawn at 5.0 s (accept reasonable tangent)
	$F = ma = \text{weight} - F_v$ $1.5(a) = 1.5(9.81) - F_v$
	$F_v = 11.2 - 11.9 \text{ N}$
	Alternative solution using formula for F_v
	At 16.0 s, $v = 35.0 \text{ m s}^{-1}$ and object is at terminal velocity. $mg = F_v$ Assuming $F_v = kv^n$, $1.5(9.81) = k35.0^n$
	At 5.0 s, $v = 29.0 \text{ m s}^{-1}$. $F = ma = mg - F_v$ $1.5(a) = 1.5(9.81) - k29.0^n$ Solving, $k = 0.198$ and $n = 1.21$
	$F_v = 0.198(29.0)^{1.21} = 11.6 \text{ N}$
(b)(i)	Assuming block A moves down the plane, $(3.0)(9.81)\sin 40^\circ - T = (3.0)a$ $T - (4.0)(9.81)\sin 20^\circ = (4.0)a$
	Solving both equations, $a = 0.785 \text{ m s}^{-2}$ (3sf)
	$(3.0)(9.81)\sin 40^\circ - T = (3.0)(0.785)$ $T = 16.6 \text{ N}$ (3sf)
(b)(ii)	Loss in $\text{GPE}_A = \text{Gain in } \text{GPE}_B + \text{Gain in } \text{KE}_{A+B} + \text{Work done against friction}$ $m_A gh_A = m_B gh_B + \frac{1}{2}(m_A + m_B)v^2 + fs$
	$(3.0)(9.81)(2.0\sin 40^\circ) = (4.0)(9.81)(2.0\sin 20^\circ) + \frac{1}{2}(3.0 + 4.0)v^2 + (4.0)(2.0)$
	$v = 0.925 \text{ m s}^{-1}$ (3sf)
	Alternative solution using dynamics and kinematics
	$(3.0)(9.81)\sin 40^\circ - (4.0)(9.81)\sin 20^\circ - 4.0 = (7.0)a$ $a = 0.214 \text{ m s}^{-1}$
	$v^2 = u^2 + 2as$ $v^2 = 0 + 2(0.214)(2.0)$
	$v = 0.925 \text{ m s}^{-1}$ (3sf)

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